# Exercise 1

$\varinjlim$ is a functor.

# Exercise 2

(1) Let $x \in X$ and $[(U, s)] \in \ker \varphi_x$. Then $[(U, \varphi_U(s))] = 0$ in $\mathcal{G}_x$, i.e. there exists an open neighbourhood $x \in V \subseteq U$ s.t. $0 = \varphi_U(s)|_V = \varphi_V(s|_V)$. But $\varphi_V$ is injective by assumption, so $s|_V = 0$. Hence $[(U, s)] = [(V, s|_V)] = 0$ and $\varphi_x$ is injective. Since $x$ was arbitrary, we conclude that $\varphi$ is injective.

For a counterexample of the reverse implication, let $|X| = 2$ with the discrete topology, let $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$. Since all restrictions are trivial, this is clearly a presheaf. Furthermore, for any $x \in X$, $\{x\}$ is final in the directed system $U \ni x$, so $\mathcal{F}_x = \mathcal{F}(\{x\}) = 0$. Now let $\mathcal{G} = 0$ on $X$ and consider the unique morphism $\mathcal{F} \to \mathcal{G}$ (again it is obvious that this is a morphism since all restrictions are trivial). Since all stalks are trivial, this is injective, but $\mathbb{Z} = \mathcal{F}(X) \to \mathcal{G}(X) = 0$ is not.

(2) This was proven as part of theorem 1.11.

(3) Let $x \in X$ and $[(U, t)] \in \mathcal{G}_x$. Since $\varphi_U$ is surjective, we can find a preimage $s \in \mathcal{F}(U)$ of $t$. Then $\varphi_x([(U, s)]) = [(U, \varphi_U(s))] = [(U, t)]$, so $\varphi_x$ is surjective. Since $x$ was arbitrary, by definition $\varphi$ is surjective.

For a counterexample of the converse, consider the presheaves of (1) and the unique morphism $\mathcal{G} \to \mathcal{F}$. For a counterexample involving sheaves, consider $\underline{\mathbb{Z}} \to i_{x,*}\mathbb{Z} \oplus i_{y,*}\mathbb{Z}$, $x \neq y \in X$, $X$ a $T_0$-space, where $i_{x,*}A$ denotes the skyscraper sheaf, the sheafification of the presheaf $U \mapsto A$ if $x \in U$ and $U \mapsto 0$ otherwise.

# Exercise 3

(1) The restriction maps of $\ker \varphi, \operatorname{im} \varphi, \operatorname{coker} \varphi$ are restrictions/quotients of the restriction maps of $\mathcal{F}$ or $\mathcal{G}$. Hence they satisfy the same identities as the original maps.

(2) Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ an open cover. Let $s \in \ker \varphi(U)$ s.t. $s|_{U_i} = 0$ for all $i \in I$. Since $\ker \varphi \subseteq \mathcal{F}$ and $\mathcal{F}$ is a sheaf, this immediately implies $s = 0$ in $\mathcal{F}(U)$.

Now let $s_i \in \ker \varphi(U_i)$ be given, satisfying $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$. Again since $\mathcal{F}$ is a sheaf, these glue to a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$. It remains to check $s \in \ker \varphi(U)$. One computes

$$\varphi_U(s)|_{U_i} = \varphi_{U_i}(s|_{U_i}) = \varphi_{U_i}(s_i) = 0,$$

and using that $\mathcal{G}$ is a sheaf, we conclude $\varphi_U(s) = 0$.

# Exercise 1

Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover of $U$. Let $s = s_1 \oplus s_2 \in (\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$ be such that $s|_{U_i} = 0$ for all $i \in I$. By definition of the restriction map, $0 = s|_{U_i} = s_1|_{U_i} \oplus s_2|_{U_i}$ implies $s_j|_{U_i} = 0$ for $j \in \{1, 2\}$. Since this holds for all $i \in I$, and $\mathcal{F}$ and $\mathcal{G}$ are sheaves, we conclude $s_j = 0$, and thus $s = s_1 \oplus s_2 = 0$.

Now let $s_i \oplus t_i \in \mathcal{F}(U_i)$ be given, s.t. $(s_i \oplus t_i)|_{U_i \cap U_j} = (s_j \oplus t_j)|_{U_i \cap U_j}$ for all $i, j \in I$. Again, by definition of the restriction maps, this means $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ and $t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}$, hence these glue to elements $s \in \mathcal{F}(U)$, $t \in \mathcal{G}(U)$ satisfying $s|_{U_i} = s_i$ and $t|_{U_i} = t_i$. Then $(s \oplus t)|_{U_i} = s|_{U_i} \oplus t|_{U_i} = s_i \oplus t_i$, so $s \oplus t \in (\mathcal{F} \oplus \mathcal{G})(U)$ is the desired element.

# Exercise 2

(1) Let $A$ be an abelian group, $X$ a topological space, $x \in X$. Write $i : \{x\} \hookrightarrow X$ for the inclusion. Then

$$i_* \underline{A}(U) = \underline{A}(i^{-1}(U)) = \begin{cases} 0 & \text{if } i^{-1}(U) = \emptyset \Leftrightarrow x \notin U, \\ A & \text{otherwise} \end{cases} = {}^x\mathcal{F}(U),$$

so ${}^x\mathcal{F} = i_*\underline{A}$ (the restriction maps clearly agree), and the result follows from exercise 3.

(2) Let $X$ be a connected $T_1$-space, $x \neq y \in X$ and $0 \neq A$ an abelian group. We are looking at the natural map $\varphi : \underline{A} \to {}^x\mathcal{F} \oplus {}^y\mathcal{F}$. Clearly $\varphi_X : A \to A \oplus A$, $a \mapsto (a, a)$ is not surjective.

First let $z \notin \{x, y\}$. Then the set of open neighbourhoods $U$ of $z$ not containing $x, y$ is cofinal in the set of all such neighbourhoods, so

$$({}^x\mathcal{F} \oplus {}^y\mathcal{F})_z = \varinjlim_{U \ni z} ({}^x\mathcal{F} \oplus {}^y\mathcal{F})(U) \cong \varinjlim_{U \ni z; x, y \notin U} ({}^x\mathcal{F} \oplus {}^y\mathcal{F})(U) = \varinjlim_{U \ni z; x, y \notin U} 0 = 0,$$

and the map $\underline{A}_z \to 0$ must be surjective.

Now let $z \in \{x, y\}$, wlog $z = x$. Similarly to above, we can compute $\varphi_x$ as the limit over the cofinal set of open neighbourhoods $U$ of $x$ not containing $y$. But on any such $U$, $\varphi_U : A \to A \oplus 0$ is the identity by definition, hence also $\varphi_x = \mathrm{id}_A : A \to A$.

# Exercise 3

$f_*\mathcal{F}$ is clearly a presheaf with the restrictions from $\mathcal{F}$. Let $U \subseteq Y$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover. Let $s \in f_*\mathcal{F}(U)$ satisfy $s|_{U_i} = 0$ for all $i \in I$. Note that $f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U_i)$ is still an open cover since $f$ is continuous. So $s \in \mathcal{F}(f^{-1}(U))$ is an element which is 0 when restricted to any open in that cover, hence $s = 0$ since $\mathcal{F}$ is a sheaf.

Now let $s_i \in f_*\mathcal{F}(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ be given. Again, since $f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U_i)$ is an open cover, these glue to an element of $\mathcal{F}(f^{-1}(U)) = f_*\mathcal{F}(U)$ satisfying $s|_{U_i} = s_i$, as desired.

# Exercise 4

(1) Since $IJ \subseteq I \cap J$, by the lecture we have $V(I \cap J) \subseteq V(IJ) = V(I) \cup V(J)$. Conversely, since $I \cap J \subseteq I$, one gets $V(I) \subseteq V(I \cap J)$ and symmetrically $V(J) \subseteq V(I \cap J)$, hence also $V(I) \cup V(J) \subseteq V(I \cap J)$.

(2) We have

$$\mathfrak{p} \in V\left(\sum_\alpha I_\alpha\right) \iff \sum_\alpha I_\alpha \subseteq \mathfrak{p} \iff I_\alpha \subseteq \mathfrak{p},\ \forall\alpha$$
$$\iff \mathfrak{p} \in V(I_\alpha),\ \forall\alpha \iff \mathfrak{p} \in \bigcap_\alpha I_\alpha.$$

# Exercise 5

First assume there exists $0, 1 \neq e \in A$ with $e^2 = e$. Then we claim that $\operatorname{Spec} A = V(e) \sqcup V(1-e)$: By exercise 4, $V(e) \cup V(1-e) = V(e(1-e)) = V(0) = \operatorname{Spec} A$, and $V(e) \cap V(1-e) = V((e)+(1-e)) = V(1) = \emptyset$. Also $V(e) \neq \emptyset$, since $e$ is a zero-divisor, hence not a unit. Since also $(1-e)^2 = 1-e$, the same argument applies for $V(1-e)$ and the above is a non-trivial decomposition of $\operatorname{Spec} A$ into two clopen subsets.

For the converse, first note that $\operatorname{Spec} A$ is quasi-compact. Indeed, it suffices to consider coverings of the form $\operatorname{Spec} A = \bigcup_i D(f_i)$. This is equivalent to $(f_i)_i = 1$, but this means $1$ is generated by finitely many of the $f_i$.

Now suppose $\operatorname{Spec} A = V(I) \sqcup V(J)$ is a non-trivial decomposition. By the same arguments as above, we have $\sqrt{0} \supseteq IJ$ and $1 = \sqrt{I+J} = I + J$. Since $V(I), V(J)$ are closed subsets of $\operatorname{Spec} A$, they are quasi-compact as well. So we can write $V(J) = \operatorname{Spec} A \setminus V(I) = \bigcup_i D(f_i)$ for finitely many $i$. Then $V(I) = V(f_i)_i$ and wlog we may assume that $I, J$ are finitely generated.

Then $IJ$ is finitely generated as well, and $IJ \subseteq \sqrt{0}$ implies $(IJ)^N = 0$ for some $N \gg 0$ (take the sum of exponents for a generating set). Raising $I + J = 1$ to the $2N-1$-th power, we see $I^N + J^N = 1$, so write $1 = e + (1-e)$ with $e \in I^N$ and $1 - e \in J^N$. Then $e(1-e) \in (IJ)^N = 0$, i.e. $e^2 = e$.

## Exercise 1

Let $\operatorname{Spec} A = Z_1 \cup Z_2$ with $Z_1, Z_2$ closed. Suppose wlog that $(0) \in Z_1$. Then $Z_1 \supseteq \overline{\{(0)\}} = \operatorname{Spec} A$.

## Exercise 2

(1) Last week.

(2) If $A = A_1 \times A_2$, then $e = (1, 0)$ is a nontrivial idempotent, so the claim follows from (1). Conversely, by (1) we have a nontrivial idempotent $e^2 = e \in A$. Let $\pi : A \to A/eA \times A/(1-e)A$ be the natural map. If $x \in \ker \pi$, then $x \in eA \cap (1-e)A = (e(1-e))A = 0$. Let $(a+eA, b+(1-e)A) \in A/eA \times A/(1-e)A$ be given. Then $\pi(a(1-e)+be) = (a(1-e)+eA, be+(1-e)A) = (a+eA, b+(1-e)A)$. So $\pi$ is an isomorphism.

(3) Let $(A, \mathfrak{m})$ be local and assume $A$ contains a non-trivial idempotent $e$. Then $e$ is in particular a zero-divisor, hence not a unit, and the same applies to $1-e$. Hence $e, 1-e \in \mathfrak{m}$, but then also $1 = e + (1-e) \in \mathfrak{m}$, contradiction.

## Exercise 3

(1) The morphism $\pi : A \to S^{-1}A$ has the following universal property: For every morphism $f : A \to B$ such that $f(S) \subseteq B^\times$, there exists a unique morphism $\widetilde{f} : S^{-1}A \to B$ with $f = \widetilde{f} \circ \pi$.

(2) For the universal property to work, we only have to check that every element of $\{1, f, \dots, f^n, \dots\}$ is invertible in $A_h$. Clearly it suffices to do this for $f$. Since $D(h) \subseteq D(f) \iff V(f) \subseteq V(h)$, by Lemma 2.2 $h \in \sqrt{(f)}$, i.e. $h^m = af$ for some $a \in A$ and $m > 0$. Since $h$ is invertible in $A_h$, so is $f$, and the result follows.

(3) These are exactly the restriction maps $\mathcal{O}(D(f)) \to \mathcal{O}(D(h))$.

## Exercise 4

(1) We show more generally:

**Lemma 1.** *Let $\mathcal{C}$ be a category with direct limits, $(I, \leq)$ a directed set indexing a directed system $(A_i)_{i\in I}$, $(\varphi_{ij} : A_i \to A_j)_{i<j}$ in $\mathcal{C}$. Let $\iota_i : A_i \to \varinjlim_{i\in I} A_i =: A$ be its direct limit. Let $J \subseteq I$ be a* cofinal *subset, that is, for every $i \in I$ there exists $j \in J$ with $i \leq j$. Then $J$ is directed and $(A_j)_{j\in J}, (\varphi_{ij})_{i<j;i,j\in J}$ is a directed system; write $\iota'_j : A_j \to \varinjlim_{j\in J} A_j =: A'$ for the direct limit of this sub-directed system. Then there exists a unique isomorphism $\varphi : A' \to A$ such that $\iota_j = \varphi \circ \iota'_j$ for every $j \in J$.*

*Proof.* $(\iota_j : A_j \to A)_{j\in J}$ satisfy $\iota_{j'} = \iota_j \circ \varphi_{jj'}$ by definition, so they induce a unique morphism $\varphi : A' \to A$ such that $\iota_j = \varphi \circ \iota'_j$. It remains to construct an inverse morphism. For $i \in I$ choose $j \in J$ with $i \leq j$, and set $\psi_i := \iota'_j \circ \varphi_{ij} : A_i \to A'$. Note that $\psi_i$ does not depend on the choice of $j$: Let $j' \in J$ be another element with $i \leq j'$, since $J$ is directed we may assume $j \leq j'$. (Otherwise pick $j'' \in J$ with $j \leq j''$ and $j' \leq j''$ and apply the result to both of these.) Then $\iota'_{j'} \circ \varphi_{ij'} = \iota'_{j'} \circ \varphi_{jj'} \circ \varphi_{ij} = \iota'_j \circ \varphi_{ij}$. Now let $i \leq k \in I$, and choose $J \ni j \geq k$. Then $\psi_k \circ \varphi_{ik} = \iota'_j \circ \varphi_{kj} \circ \varphi_{ik} = \iota'_j \circ \varphi_{ij} = \psi_i$, so the $(\psi_i)_{i\in I}$ induce a morphism $\psi : A \to A'$ with $\psi_i = \psi \circ \iota_i$.

To show $\psi \circ \varphi = \mathrm{id}_{A'}$, by uniqueness it suffices to show $\psi \circ \varphi \circ \iota'_j = \iota'_j$ for every $j \in J$. Indeed, $\psi \circ \varphi \circ \iota'_j = \psi \circ \iota_j = \psi_j = \iota'_j \circ \varphi_{jj} = \iota'_j$. Similarly, $\varphi \circ \psi \circ \iota_i = \varphi \circ \psi_i = \varphi \circ \iota'_j \circ \varphi_{ij} = \iota_j \circ \varphi_{ij} = \iota_i$ for every $i \in I$, where $i \leq j \in J$, implies $\varphi \circ \psi = \mathrm{id}_A$. $\square$

Back to the exercise: The directed system in question is $(\{U \subseteq X \text{open} \mid x \in U\}, \supseteq)$ so by definition of a base of a topological space it is clear that $(\{U \in \mathcal{B} \mid x \in U\}, \supseteq)$ is a sub-directed set.

(2) From the proposition and (1), we immediately have

$$A_\rho \cong \mathcal{O}_{\operatorname{Spec} A,\rho} = \varinjlim_{U \ni \rho} \mathcal{O}_{\operatorname{Spec} A}(U) \cong \varinjlim_{D(f) \ni \rho} \mathcal{O}_{\operatorname{Spec} A}(D(f)) \cong \varinjlim_{D(f) \ni \rho} A_f.$$

By commutative algebra alone, for $f \notin \rho$, $f$ is invertible in $A_\rho$ by definition, so the universal property yields a unique morphism $\varphi_f$ s.t. $\pi = \varphi_f \circ \pi_f$, where $\pi : A \to \mathbb{A}_\rho$, $\pi_f : A \to A_f$ are the canonical maps. Write further $\varphi_{ij} : A_f \to A_h$ for the maps constructed in exercise 3(2). The $(\varphi_{fh})_{D(f) \supseteq D(h)}$ is clearly a directed system, and $\varphi_h \circ \varphi_{fh}$ satisfies the universal property of $\varphi_f$, hence they are equal. Thus they induce a morphism $\varphi : \varinjlim_{D(f) \ni \rho} A_f \to A_\rho$, given by $[\frac{a}{f^n} \in A_f] \mapsto \frac{a}{f^n} \in A_\rho$. $\varphi$ is clearly surjective, let $\alpha = [\frac{a}{f^n} \in A_f] \in \ker \varphi$. Then $as = 0$ for some $s \notin \rho$, hence $\frac{a}{f^n} = \frac{as}{f^n s} = 0$ in $A_{f^n s}$, i.e. $\alpha = 0$.

# Exercise 1

(1) $f^{-1}(D(h)) = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi^{-1}(\mathfrak{p}) \in D(h)\} = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi(h) \notin \mathfrak{p}\} = D(\varphi(h))$.

(2) Let $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$. Using (1), we have a commutative diagram

$$\begin{tikzcd}
B \arrow[r, "\cong"] \arrow[d] & \mathcal{O}_X(X) \arrow[d] & \mathcal{O}_Y(Y) \arrow[l, "f^\sharp_Y"'] \arrow[d] & A \arrow[l, "\cong"] \arrow[d] \arrow[lll, bend right, "\varphi"'] \\
B_{\varphi(h)} \arrow[r, "\cong"] & \mathcal{O}_X(D(\varphi(h))) & \mathcal{O}_Y(D(h)) \arrow[l, "f^\sharp_{D(h)}"'] & A_h \arrow[l, "\cong"] \arrow[lll, dashed, bend left, "\varphi_h"]
\end{tikzcd}$$

where the vertical arrows are localizations/restrictions and the isomorphisms on the left and right are proposition 2.11. The diagram of solid arrows commutes: The central square by the definition of a morphism of sheaves, the outer squares by the proof of proposition 2.11, the upper part by proposition 2.18. Hence the composition along the bottom row satisfies the universal property of $\varphi_h$, so they agree. In other words, the map $f^\sharp_D(h)$ is, up to identifications of sections with $A_h$ resp. $B_{\varphi(h)}$, exactly the localization morphism $\varphi_h$.

# Exercise 2

Follows directly from proposition 1.11 and exercise 3.4(1).

# Exercise 3

(1) The localization map induces an inclusion-preserving bijection between the prime ideals of $S^{-1}A$ and the prime ideals of $A$ not intersecting $S$.

(2) $\operatorname{im} f = \{\mathfrak{p} \in \operatorname{Spec} A \mid \varphi(\mathfrak{p}) \in \operatorname{Spec} A_g\} \overset{(1)}{=} \{\mathfrak{p} \in \operatorname{Spec} A \mid g \notin \mathfrak{p}\} = D(g)$. In the same way one shows $f(D(h)) = D(h) \cap D(g)$, so $f$ is an open continuous map, i.e. a homeomorphism onto its image.

(3) By (2) we have a map $(f, f^\sharp) : \operatorname{Spec} A_g \to (D(g), \mathcal{O}_{\operatorname{Spec} A}|_{D(g)})$, which is a homeomorphism on topological spaces. It remains to check that $f^\sharp$ is an isomorphism, for which by exercise 2 it suffices to consider principal opens $D(h)$, $D(h) \subseteq D(g)$. Then by exercise 1, $f^\sharp_{D(h)}$ is canonically identified with the localization map $A_h \to (A_g)_h$, which is an isomorphism since they satisfy the same universal property.

# Exercise 4

By proposition 2.25, $\operatorname{Hom}_{\mathtt{Sch}}(\_, \operatorname{Spec} A)$ is a sheaf on $X$, and since $\operatorname{Hom}_{\mathtt{Ring}}(A, \_)$ preserves limits, so is $\operatorname{Hom}_{\mathtt{Ring}}(A, \mathcal{O}_X(-))$. Checking that the given morphism between those sheaves is an isomorphism can be done on a basis by exercise 2, so by lemma 2.22 it suffices to consider $X$ affine, which was done in proposition 2.18.

# Exercise 1

(1) $\mathcal{O}_{X,\xi} \cong A_{(0)}$ is the localization at $A \setminus 0$, i.e. the fraction field.

(2) Let $U$ be open. Since $\xi \in U$, by definition of the direct limit, we have morphisms $\varphi_U : \mathcal{O}_X(U) \to \mathcal{O}_{X,\xi} \to \operatorname{Frac} A$ such that $\varphi_V \circ \rho_{UV} = \varphi_U$ whenever $V \subseteq U$. This exactly says that the $(\varphi_U)_U$ form a morphism of sheaves $\varphi : \mathcal{O}_X \to \mathcal{F}$. We can check injectivity on the stalks, so let $\mathfrak{p} \in X$. Then by construction of the structure sheaf, $\mathcal{O}_{X,\mathfrak{p}} \cong A_\mathfrak{p} \to A_{(0)} \cong \mathcal{F}_\mathfrak{p}$ is the map induced by localization, which is injective since $A$ is a domain.

# Exercise 2

(1) First suppose $X$ is reduced, let $x \in X$. Since $x$ lies in some affine and the stalk can be computed on it by lemma 1, wlog let $X = \operatorname{Spec} A$ be affine. Now suppose $(\frac{a}{s})^n = 0$ for some $\frac{a}{s} \in A_\mathfrak{p}$, $n \geq 1$. Then $s'a^n = 0$ for some $s' \notin \mathfrak{p}$, thus $(s'a)^n = s'^{n-1}(s'a^n) = 0$. Since $\mathcal{O}_X(X) = A$ is reduced by assumption, $s'a = 0$, hence $\frac{a}{s} = 0$ and $A_\mathfrak{p}$ is reduced.

Conversely, suppose all stalks $\mathcal{O}_{X,x}$ are reduced, and let $U \subseteq X$ be open. Suppose $s^n = 0$ for some $s \in \mathcal{O}_X(U)$. For every $x \in U$, then $[(U, s^n)] = [(U, s)]^n = 0 \in \mathcal{O}_{X,x}$, so $[(U, s)] = 0$, which means $s|_{V_x} = 0$ for some open $x \in V_x \subseteq U$. But the $(V_x)_{x \in U}$ cover $U$, so $s = 0$ by the sheaf property.

(2) By the proof of (1), $A$ reduced implies that all stalks of $\operatorname{Spec} A$ are reduced, so we conclude with (1). The converse follows immediately from (1).

# Exercise 3

(1) First suppose $\xi = \operatorname{Nil} A$ is prime and let $X = X_1 \cup X_2$ with $X_1, X_2$ closed. Wlog let $\xi \in X_1$, then $X_1 \supseteq \overline{\{\xi\}} = X$, since the nilradical is contained in every prime ideal. Conversely, suppose $X$ is irreducible and let $p, q \in A$ with $pq \in \operatorname{Nil} A$. Then $V(p) \cup V(q) = V(pq) = V(0) = X$, so wlog $V(p) = X$, i.e. $p \in \operatorname{Nil} A$.

(2) If $X$ is integral, then in particular so is $\mathcal{O}_X(X) = A$. Conversely, if $A$ is integral, then by exercise 1 $\mathcal{O}_X(U) \subseteq \operatorname{Frac} A$ for all $U \subseteq X$ open, and subrings of integral domains are integral domains.

# Exercise 4

Let $k$ be a field and consider $X = \operatorname{Spec}(k \times k) = \operatorname{Spec} k \sqcup \operatorname{Spec} k$. Then the stalks are just $k$, which are integral domains, but $k \times k$ is not.

# Exercise 1

Let $\mathfrak{p} \subseteq A$ be a prime ideal. Since $X_i^2 = 0 \in \mathfrak{p}$, we must have $X_i \in \mathfrak{p}$ for all $i \in \mathbb{N}$, hence $\mathfrak{m} := (X_1, X_2, \ldots) \subseteq \mathfrak{p}$. But this is already a maximal ideal, since $A/\mathfrak{m} \cong K$ is a field. Thus $\operatorname{Spec} A = \{\mathfrak{m}\}$ is a singleton, and hence in particular Noetherian. But $A$ is not a Noetherian ring due to the strictly increasing chain $(X_1) \subsetneq (X_1, X_2) \subsetneq (X_1, X_2, X_3) \subsetneq \ldots$ of ideals in $A$.

# Exercise 2

(1) First let $X$ be Noetherian, $U \subseteq X$ be open, and $U = \bigcup_{i \in I} U_i$ be an open cover. Let $U_0$ be any of the $U_i$. We recursively define a sequence $(U_n)_{n \geq 0}$ as follows: If $\bigcup_{j=0}^n U_j = U$, then we have found a finite subcover and are done. Otherwise, there is $x \in U \setminus \bigcup_{j=0}^n U_j$, then let $U_{n+1}$ be some $U_i$ containing $x$. This procedure cannot proceed indefinitely, since then $(\bigcup_{j=0}^n U_j)_{n \geq 0}$ would be a strictly ascending chain of opens. Conversely, if any open in $X$ is quasi-compact and $U_0 \subseteq U_1 \subseteq \ldots$ is an ascending chain of opens, let $U := \bigcup_j U_j$ Since $U$ is quasi-compact by assumption, there exists a finite subcover $U = \bigcup_{k=1}^n U_{j_k}$. Thus the chain stabilizes at $\max(j_1, \ldots, j_n)$.

(2) Follows directly from (1) since any open in $U$ is also open in $X$.

# Exercise 3

**Lemma 2.** *Let $\mathcal{F}$ be a presheaf and $\mathcal{G}$ be a sheaf on a topological space $X$, and let $\mathcal{B}$ be a basis of $X$. Let $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$, $B \in \mathcal{B}$ be a collection of morphisms, such that whenever $B' \subseteq B$, $B', B \in \mathcal{B}$, then $\varphi_{B'} \circ \rho_{BB'} = \rho_{BB'} \circ \varphi_B$, where the $\rho$'s are the restriction maps of $\mathcal{F}, \mathcal{G}$. Then there exists a unique morphism $\varphi : \mathcal{F} \to \mathcal{G}$ with $\varphi(B) = \varphi_B$ for all $B \in \mathcal{B}$.*

*Proof.* Let $U \subseteq X$ be open, and $U = \bigcup_{i \in I} B_i$ be a cover by basis elements. Let $s \in \mathcal{F}(U)$. The elements $\varphi_{B_i}(s)$ glue by assumption to a unique element $\varphi(U)(s) \in \mathcal{G}(U)$, since $B_i \cap B_j$ is again covered by basis elements. We have to check that this definition of $\varphi(U)$ is independent of the given cover, then everything follows immediately.

So let $U = \bigcup_{j \in J} B_j$ be another cover, and denote by $\varphi'(U)$ the associated morphism. Let $B_i \cap B_j = \bigcup_k B_{ijk}$ for all $i, j$, so that $U = \bigcup_{i,j,k} B_{ijk}$ is a common refinement of the two covers. Then

$$\begin{aligned} \varphi(U)(s)|_{B_{ijk}} &= \varphi(U)(s)|_{B_i}|_{B_{ijk}} = \varphi_{B_i}(s)|_{B_{ijk}} = \varphi_{B_{ijk}}(s) \\ &= \varphi_{B_j}(s)|_{B_{ijk}} = \varphi'(U)(s)|_{B_j}|_{B_{ijk}} = \varphi'(U)(s)|_{B_{ijk}}, \end{aligned}$$

hence $\varphi(U) = \varphi'(U)$ by the sheaf condition. □

(1) Clearly $X_{\text{red}}$ is a locally ringed space.

First let $X = \operatorname{Spec} A$ be an affine scheme. We show that $X_{\text{red}} \cong \operatorname{Spec} A_{\text{red}}$: Since $\operatorname{Nil} A$ is the intersection of all prime ideals of $A$, the topological map $f : \operatorname{Spec} A_{\text{red}} \to \operatorname{Spec} A$ associated to $A \mapsto A/\operatorname{Nil} A$ is a homeomorphism. Denote by $X_{\text{red}}^{\text{pre}}$ the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$ from the definition of $X_{\text{red}}$. Let $g \in A$ and set $\varphi_{D(g)} : (\mathcal{O}_X)_{\text{red}}^{\text{pre}} = (A_g)_{\text{red}} \to (A_{\text{red}})_g \cong \mathcal{O}_{\operatorname{Spec} A_{\text{red}}} \cong f_*\mathcal{O}_{\operatorname{Spec} A_{\text{red}}}$ to be the natural isomorphism (localization commutes with quotients). By naturality, this isomorphism commutes with restrctions, so by lemma 2 we obtain a morphism of presheaves $\varphi : (\mathcal{O}_X)_{\text{red}}^{\text{pre}} \to f_*\mathcal{O}_{\operatorname{Spec} A_{\text{red}}}$. Since its codomain is a sheaf, by the universal property of sheafification this induces a morphism $f^\sharp : (\mathcal{O}_X)_{\text{red}} \to f_*\mathcal{O}_{\operatorname{Spec} A_{\text{red}}}$ of sheaves. Since all the $\varphi_B$ were isomorphisms, so are the induced morphisms of stalks $f_x^\sharp = \varphi_x$, $x \in X$. Therefore $(f, f^\sharp) : \operatorname{Spec} A_{\text{red}} \to X_{\text{red}}$ is an isomorphism of schemes.

Now let $X$ be arbitrary, and $\operatorname{Spec} A \cong U \subseteq X$ be an affine open subset. Then $(\mathcal{O}_X)_{\text{red}}|_U = (\mathcal{O}_U)_{\text{red}} \cong \operatorname{Spec} A_{\text{red}}$, so $X$ is locally isomorphic to affine schemes, i.e. a scheme. Also, it is clear that $X$ is reduced, since that can be checked on stalks, and $(A_{\text{red}})_{\mathfrak{p}} \cong (A_{\mathfrak{p}})_{\text{red}}$ is reduced, for any $\mathfrak{p} \in \operatorname{Spec} A \subseteq X$.

To construct $i$, take the identity on topological spaces, and the obvious map $\mathcal{O}_X(U) \to \mathcal{O}_X(U)_{\text{red}}$, followed by the sheafification map. On an affine open subset, this is exactly the map associated to $A \mapsto A_{\text{red}}$.

(2) On topological spaces, take $g = f$, then $f = i \circ g$ is clear. Consider the diagram

$$\begin{tikzcd}
f_*\mathcal{O}_X & \mathcal{O}_Y \arrow[l, "f^\sharp"'] \arrow[d, "\text{red}"] \arrow[dl, "i^\sharp"] \\
(\mathcal{O}_Y)_{\text{red}} \arrow[u, dashed, "g^\sharp"] & (\mathcal{O}_Y)^{\text{pre}}_{\text{red}} \arrow[l, "(-)^+"] \arrow[ul, dashed]
\end{tikzcd}$$

Let $U \subseteq Y$ be open. Since $f_*\mathcal{O}_X(U)$ is reduced, $f^\sharp_U$ factors through $\mathcal{O}_Y(U)_{\text{red}}$. This gives a morphism of presheaves $h : (\mathcal{O}_Y)^{\text{pre}}_{\text{red}} \to f_*\mathcal{O}_X$, hence by the universal property of sheafification a morphism $g^\sharp(\mathcal{O}_Y)_{\text{red}} \to f_*\mathcal{O}_X$ with $g^\sharp \circ i^\sharp = g^\sharp \circ (-)^+ \circ \text{red} = h \circ \text{red} = f^\sharp$ as required.

# Exercise 4

(1) Clearly $0 \in I \subseteq I^S$. Let $a, a' \in I^S$, with $sa, s'a' \in I$ for suitable $s, s' \in S$. Then $ss'(a + a') = s'(sa) + s(s'a') \in I$, hence $a + a' \in I^S$. For $r \in A$ we have $s(ra) = r(sa) \in I$, so $ra \in I^S$.

(2) Let $a \in I^S$, $sa \in I$ for $s \in S$. Then $\varphi(a) = \frac{a}{1} = \frac{sa}{s} \in S^{-1}I$. Conversely, let $a \in \varphi^{-1}(S^{-1}I)$, say $\frac{a}{1}\varphi(a) = \frac{i}{s}$. Then there is $s' \in S$ with $s'(as - i) = 0$, i.e. $s'sa = s'i \in I$, so $a \in I^S$.

(3) Let

$$\begin{aligned}
\Phi : \{I \subseteq A \text{ ideal} \mid I^S = I\} &\leftrightarrows \{\text{ideals } J \subseteq S^{-1}A\} : \Psi \\
I &\mapsto S^{-1}I \\
\varphi^{-1}(J) &\leftarrow\!\shortmid J
\end{aligned}$$

$\Psi$ is well-defined since if $a \in (\varphi^{-1}(J))^S$, say $sa \in \varphi^{-1}(J)$, then $\frac{sa}{1} \in J$. Since $s$ is invertible in $S^{-1}A$, it follows that $\frac{a}{1} \in J$ and hence $a \in \varphi^{-1}(J)$.

Now by (2), $\Psi(\Phi(I)) = \varphi^{-1}(S^{-1}I) = I^S$ shows $\Psi \circ \Phi = \text{id}$, so $\Phi$ is injective. Let $J \subseteq S^{-1}A$ be an ideal, $\frac{a}{s} \in J$. As above, it follows that $a \in \varphi^{-1}(J)$ and hence $\frac{a}{s} \in S^{-1}(\varphi^{-1}(J))$, so $J \subseteq S^{-1}(\varphi^{-1}(J))$. Conversely, let $\frac{a}{s} \in S^{-1}(\varphi^{-1}(J))$ with $\varphi(a) \in J$. Then $\frac{a}{s} = \frac{1}{s}\varphi(a) \in J$. Therefore, $\Phi \circ \Psi = \text{id}$ as well.

# Exercise 1

(1) By a previous exercise, $X \to \operatorname{Spec} A$ is equivalent to a ring morphism $A \to \mathcal{O}_X(X)$, and composition with the restriction maps $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$ gives a unique $A$-algebra structure on the sheaf $\mathcal{O}_X$.

(2) There are unique maps $\mathbb{Z} \to \mathcal{O}_X(U)$ for any $U \subseteq X$ open. By uniqueness, they form a sheaf of $\mathbb{Z}$-algebras, hence by (1) $X$ is a sheaf over $\mathbb{Z}$ in a unique way.

# Exercise 2

(1) We apply lemma 3.19 with the property $\mathcal{P}(\operatorname{Spec} B) = B$ is a finitely generated $A$-algebra. If $B$ is finitely generated and $f \in B$, then $B_f \cong B[X]/(Xf - 1)$ is finitely generated over $B$, hence also over $A$. Now let $(b_1, \dots, b_n) = 1$ and assume that all $B_{b_i}$ are finitely generated $A$-algebras. Write $1 = \sum_{i=1}^n c_i b_i$. For fixed $i$, let $\frac{a_{i1}}{b_i^{k_1}}, \dots, \frac{a_{im}}{b_i^{k_m}}$ be a generating set, and set $N_i = \{a_{i1}, \dots, a_{im}\}$. Let $S = \bigcup_{i=1}^n N_i \cup \{b_1, \dots, b_n\} \cup \{c_1, \dots, c_n\}$. We show that $B$ is generated by $S$. Let $\varphi : C = A[S] \hookrightarrow B$ the inclusion; then $\varphi_{b_i} : C_{b_i} \hookrightarrow B_{b_i}$ is surjective, since $C_{b_i}$ contains $\frac{1}{b_i} N_i$. Hence $C_{b_i} \cong B_{b_i}$ for all $b_i$.

Now a standard local-global argument shows $C = B$: Let $M = B/C$, then $M_{b_i} \cong B_{b_i}/C_{b_i} = 0$. Let $m \in M$. Since $\frac{m}{1} = 0 \in M_{b_i}$, there exists $k \in \mathbb{N}$ with $b_i^k m = 0$ for all $i = 1, \dots, n$. Therefore $(m) = (m)(b_1^k, \dots, b_n^k) = (b_1^k m, \dots, b_n^k m) = 0$, hence $M = 0$ and $B = C$ is finitely generated as an $A$-algebra.

(2) Clear.

# Exercise 3

Let $(i, i^\sharp) : \operatorname{Spec} K \to X$ be a morphism of schemes. Since $\operatorname{Spec} K$ has one point, $i(\operatorname{Spec} K) = \{x\}$ is a singleton. Hence $i_* \mathcal{O}_{\operatorname{Spec} K}$ is the skyskraper sheaf with value $K$ at $x$, so the map $i^\sharp : \mathcal{O}_X \to i_* \mathcal{O}_{\operatorname{Spec} K}$ is uniquely determined by its induced map of stalks $i_x : \mathcal{O}_{X,x} \cong (i_* \mathcal{O}_{\operatorname{Spec} K})_x = K$, since we must have $i_U^\sharp(s) = i_x([U, s])$ for any $U \subseteq X$ open and $s \in \mathcal{O}_X(U)$. Now $i_x$ is local by definition, so $i_x(\mathfrak{m}_x) = 0$, so $i_x$ factors through $\alpha : k(x) \to K$. Therefore we have constructed a map from $X(K)$ to pairs $(x, \alpha)$ as desired. Conversely, given $(x, \alpha)$, set $i((0)) = x$ and $i_U^\sharp(s) = \alpha(\overline{s})$ for all $U \subseteq X$ open and $s \in \mathcal{O}_X(U)$. This clearly gives an elemet of $\operatorname{Spec} K \to X$. These two constructions are obviously inverse to each other.

# Exercise 4

(1) The structure morphism $X \to \operatorname{Spec} k$ induces on stalks a field extension $k \to \mathcal{O}_{X,x} \to k(x)$. By exercise (3), if $(x, \alpha) \in X(K)$, then $k \to k(x) \to k$ is the identity, so both maps have to be isomorphisms. The converse is clear.

(2) Clear from (1), since $\mathcal{O}_{X,x} \cong \mathcal{O}_{U,x}$.

(3) By proposition 2.18, we are looking for morphisms $\varphi : A = k[X_1, \dots, X_n]/(f_1, \dots, f_m) \to k$ such that precomposition with $i : k \hookrightarrow A$ is the identity. By the universal properties of polynomial rings and quotients, those maps $\varphi$ correspond exactly to tuples $x \in k^n$ with $f_i(x) = 0$ for all $i$.

(4) By (3), we are looking for real solutions of $0 = X^2 + Y^2 + 1 \geq 1$, so $X(\mathbb{R}) = \emptyset$. Let $x = (X, Y^2 + 1) \in X$. Then $(\mathbb{R}[X, Y]/(X^2 + Y^2 + 1))/x \cong R[Y]/(Y^2 + 1) \cong \mathbb{C}$ as $\mathbb{R}$-algebras via $Y \mapsto \pm i$, so we get two morphism $i : \operatorname{Spec} \mathbb{C} \to X$ of $\mathbb{R}$-schemes.

# Exercise 1

First let $x$ be a closed point, with corresponding maximal ideal $\mathfrak{m}$. Then $A/\mathfrak{m}$ is already a field, so $k(x) = A/\mathfrak{m}$, which is finitely generated by $\{\overline{1}\}$, hence algebraic by Zariski's Lemma. Conversely, suppose $K = \operatorname{Frac}(A/\mathfrak{p})/k$ is algebraic. Then $A/\mathfrak{p}$ is finitely generated over $k$, and every generator is algebraic, hence $A/\mathfrak{p}$ is finite over $k$. But every integral domain finite over a field is itself a field, since then multiplication by any nonzero element is injective, hence surjective.

# Exercise 2

We have to prove that every open contains a closed point. It clearly suffices to do this for a basis, so let $\emptyset \neq \operatorname{Spec} A \subseteq X$. Since $A \neq 0$, $\operatorname{Spec} A$ has a maximal ideal/closed point $x$. By exercise 1, this remains a closed point in $X$, since $k(x)$ does not depend on the neighbourhood of $x$.

# Exercise 3

$\operatorname{Spec} k[\![t]\!] = \{0, (t)\}$ and hence $\{0\}$ is an open subset without a closed point. Thus by exercise 2, $X$ is not locally of finite type over $k$, i.e. not a finitely generated $k$-algebra by sheet 7, exercise 2.

# Exercise 4

(1) Write $d := \deg f$. Let $\mathfrak{p} \in D_+(f)$ and $x \in \mathfrak{p}$ homogeneous of degree $n$. Then $\frac{x^d}{f^n} \in \mathfrak{p}B_f \cap B_{(f)} = \theta(\mathfrak{p})$, hence by definition $x \in \psi(\theta(\mathfrak{p}))_n \subseteq \psi(\theta(\mathfrak{p}))$ . Since $\mathfrak{p}$ is homogeneous, this implies $\mathfrak{p} \subseteq \psi(\theta(\mathfrak{p}))$. Conversely, let $x \in \psi(\theta(\mathfrak{p}))_n$ for some $n$, i.e. $\frac{x^d}{f^n} \in \theta(\mathfrak{p})$. This implies $f^i x^d \in \mathfrak{p}$ for some $i$, hence $x \in \mathfrak{p}$ since $\mathfrak{p}$ is prime. Again by homogenity, $\psi(\theta(\mathfrak{p})) \subseteq \mathfrak{p}$ and hence $\psi \circ \theta = \mathrm{id}_{D_+(f)}$.

Conversely, let $\mathfrak{q} \in \operatorname{Spec} B_{(f)}$ and $\frac{x}{f^n} \in \mathfrak{q}$. Then $x^d \in \psi(\mathfrak{q})_{dn} \subseteq \psi(\mathfrak{q})$, so $x \in \psi(\mathfrak{q})$ and hence $\frac{x}{f^n} \in \theta(\psi(\mathfrak{q}))$. Conversely, let $\frac{x}{f^n} \in \theta(\psi(\mathfrak{q}))$. Then $f^i x \in \psi(\mathfrak{q})$ for some $i$, and since $\psi(\mathfrak{q}) \in D_+(f)$ even $x \in \psi(\mathfrak{q})_{dn}$. By definition, this means $(\frac{x}{f^n})^d = \frac{x^d}{f^{dn}} \in \mathfrak{q}$, i.e. $\frac{x}{f^n} \in \mathfrak{q}$. This shows $\theta \circ \psi = \mathrm{id}_{\operatorname{Spec} B_{(f)}}$.

(2) Let $\mathfrak{p} \in \operatorname{Proj} B$. Then $h \in \theta(\mathfrak{p})$ if and only if $g \in \psi(\theta(\mathfrak{p}))_{\deg g} \subseteq \psi(\theta(\mathfrak{p})) = \mathfrak{p}$ by definition, so $\theta(D_+(g)) = D(h)$.

# Exercise 1

We already know that $f : \operatorname{Spec} A/I \to \operatorname{Spec} A$ induces a homeomorphism onto the closed subset $V(I) \subseteq \operatorname{Spec} A$, so it remains to show that $f^\sharp : \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_{\operatorname{Spec} A/I}$ is surjective. For $h \in A$ the map $A_h \to (A/I)_{h+I} \cong A_h/I_h$ is surjective, so by sheet 4, exercise 1 so are $f^\sharp_{D(h)}$ and the claim follows.

# Exercise 2

(1) Denote the image of $f$ in $\mathcal{O}_X(X_f)$ by $\overline{f}$. For $x \in X_f$, we have $\overline{f}_x = f_x \notin m_x$, so since $\mathcal{O}_{X,x}$ is a local ring, $\overline{f}_x$ is invertible in $\mathcal{O}_{X,x}$. That means there exists an open neighbourhood $x \in U_x \subseteq X_f$ and $g^{(x)} \in \mathcal{O}_X(U_x)$ with $f|_{U_x} g^{(x)} = 1$. For $x, y \in X_f$ one checks

$$f|_{U_x\cap U_y} g^{(x)}|_{U_x\cap U_y} = (f|_{U_x} g^{(x)})|_{U_x\cap U_y} = 1|_{U_x\cap U_y} = 1 = f|_{U_x\cap U_y} g^{(y)}|_{U_x\cap U_y}$$

so from uniqueness of inverses $g^{(x)}|_{U_x\cap U_y} = g^{(y)}|_{U_x\cap U_y}$ and the $(g^{(x)})_x$ glue to an element $g \in \mathcal{O}_X(X_f)$ with $g|_{U_x} = g^{(x)}$. Thus $(fg)|_{U_x} = f|_{U_x} g^{(x)} = 1 = 1|_{U_x}$ by construction, and thus $fg = 1$ again by the sheaf condition.

(2) Clear.

(3) Let $X = \bigcup_{i=1}^n U_i$ be a finite cover by affine opens $U_i \cong \operatorname{Spec} A_i$. Let $a_i := a|_{U_i}$, $f_i := f|_{U_i}$. Then $0 = \overline{a}|_{U_i \cap X_f} = a_i|_{X_f \cap U_i} = a_i|_{D(f_i)}$, so $(f^{n_i}a)|_{U_i} = f_i^{n_i} a_i = 0$ for some $n_i \geq 0$. By the sheaf condition, $f^n a = 0$ for $n = \max\{n_i\}_i$.

Now let $\frac{a}{f^m} \in \ker \psi$. Since $f \in \mathcal{O}_X(X_f)^\times$, also $\frac{a}{1} \in \ker\psi$, so by construction $a|_{X_f} = 0$. By the previous paragraph $f^n a = 0$ for some $n \geq 0$, which means $\frac{a}{f^m} = 0$ in $\mathcal{O}_X(X)_f$, i.e. $\ker\psi = 0$.

(4) Let $U_i = \operatorname{Spec} A_i$ and $V_i = X_f \cap U_i$, $f_i = f|_{V_i}$, so that $V_i = D(f_i) \subseteq \operatorname{Spec} A_i$. Let $b \in \mathcal{O}_X(X_f)$. Then $b|_{V_i} = \frac{b_i}{f_i^{n_i}} \in (A_i)_{f_i}$ for some $b_i \in A_i$ and $n_i \geq 0$. Now let $f_{ij} = f_i|_{X_f \cap U_i \cap U_j}$, then $b_i|_{X_f\cap U_i\cap U_j} = f_{ij}^{n_i} b|_{X_f\cap U_i \cap U_j}$ and

$$f_{ij}^{n_i} b_j|_{X_f\cap U_i\cap U_j} - f_{ij}^{n_j} b_i|_{X_f\cap U_i\cap U_j} = (f_{ij}^{n_i+n_j} - f_{ij}^{n_j+n_i}) b|_{X_f\cap U_i\cap U_j} = 0.$$

Let $a_{ij} = (f_j^{n_i} b_j)|_{U_i\cap U_j} - (f_i^{n_j} b_i)|_{U_i\cap U_j}$, then the above line says $a_{ij}|_{X_f\cap U_i\cap U_j} = 0$, so by (3) and the assumption that $U_i \cap U_j$ can be covered by a finite union of affine opens it follows that $f_{ij}^{k_{ij}} a_{ij} = 0$ for some $k_{ij}$. Let $N \geq k_{ij} + n_i + n_j$ for all finitely many $i, j = 1, \ldots, r$. Then $0 = f_{ij}^{N-n_i-n_j} a_{ij} = (f_j^{N-n_j} b_j)|_{U_i\cap U_j} - (f_i^{N-n_i} b_i)|_{U_i\cap U_j}$ for all $i, j$, so the $f_i^{N-n_i} b_i$ glue to a section $a \in \mathcal{O}_X(X)$ with $a|_{U_i} = f_i^{N-n_i} b_i$.

It follows that $a|_{V_i} = f_i^{N-n_i} b_i = f_i^N b|_{V_i}$, so $a|_{X_f} = f^N|_{X_f} b$ and $\psi(\frac{a}{f^N}) = b$, i.e. $\psi$ is surjective.

# Exercise 3

Denote the map by $f$. To see that the map is well-defined, assume wlog that $\alpha_0 \neq 0$. Then $f(\alpha) = \langle X_i - \frac{\alpha_i}{\alpha_0} X_0 \rangle_{i=1,\ldots,n}$, since

$$\alpha_i X_j - \alpha_j X_i = \frac{\alpha_j}{\alpha_0}(\alpha_i X_0 - \alpha_0 X_i) - \frac{\alpha_i}{\alpha_0}(\alpha_j X_0 - \alpha_0 X_j).$$

Then $\varphi_\alpha : k[X_0, \ldots, X_n] \to k[X_0]$, $X_i \mapsto \frac{\alpha_i}{\alpha_0} X_0$ is surjective and has kernel $f(\alpha)$, so $f(\alpha)$ is prime and does not contain $k[X_0, \ldots, X_n]_+$. Furthermore, under the isomorphism $D_+(X_0) \cong \operatorname{Spec} k[X_1, \ldots, X_n]$ the ideal $f(\alpha)$ corresponds to $(X_i - \frac{\alpha_i}{\alpha_0})_i$, so $k(f(\alpha)) = k$ and $f$ is well-defined.

To see that $\varphi$ is injective, let $\alpha, \beta \in \mathbb{P}(k^n)$ with $f(\alpha) = f(\beta)$, and wlog $\alpha_0 \neq 0$. If $\beta_0 = 0$, then $X_0 \in f(\beta) \setminus f(\alpha)$, contradiction. So $\beta_0 \neq 0$, and now $f(\alpha), f(\beta)$ are the kernels of $\varphi_\alpha, \varphi_\beta$, so $\varphi_\alpha = \varphi_\beta$ and hence $\alpha = \beta$.

For surjectivity let $\mathfrak{p} \in \mathbb{P}_k^n$ with $k(\mathfrak{p}) = k$. Wlog $X_0 \notin \mathfrak{p}$. Under the isomorphism $D_+(X_0) \cong \operatorname{Spec} k[X_1, \dots, X_n]$, $\mathfrak{p}$ maps to a prime ideal with residue field $k$, so by sheet 7, exercise 4 one of the form $(X_i - a_i)_i$ for some $a_i \in k$. Then $\mathfrak{p} = (X_i - a_i X_0)_i = f([1 : a_1 : \dots : a_n])$.

# Exercise 4

Let $f \in \mathcal{O}_X(X)$. Write $f_i := f|_{D_+(X_i)} = \frac{g_i}{X_i^{n_i}} \in A[X_0, \dots, X_n]_{(X_i)}$. Since the $f_i$ agree on intersections, we have $(X_i X_j)^{m_{ij}}(X_i^{n_i} g_j - X_j^{n_i} g_i) = 0$ for some $m_{ij} \geq 0$. Multiplying the $g_i$ by suitable powers of the $X_i$, we may assume $X_i^n g_j - X_j^n g_i = 0$ for all $i, j$. But this means $X_i^n \mid g_i$, which is homogeneous of degree $n$, so $g_i = a_i X_i^n$. Now the equation reads $(a_i - a_j) X_i^n X_j^n = 0$, so $a_i = a_j =: a$ and $f_i = a$ for all $i$. By the sheaf condition, they glue to a unique element of $\mathcal{O}_X(X)$, which is clearly $a$ (under the canonical map $A \to \mathcal{O}_X(X)$).

# Exercise 1

(1) If $f^\sharp$ is injective, then so is in particular $f^\sharp_{\operatorname{Spec} A} = \varphi$. Conversely, assume that $\varphi$ is injective. Since localization is exact, then so is $\varphi_h : A_h \to B_{\varphi(h)}$, hence $f^\sharp_{D(h)} : \mathcal{O}_{\operatorname{Spec} A}(D(h)) \to \mathcal{O}_{\operatorname{Spec} B}(D(\varphi(h)))$ is injective as well. Passing to stalks, this immediately implies that $f^\sharp$ is injective.

(2) Let $\emptyset \neq D(h) \subseteq \operatorname{Spec} A$ be open. Since $h$ is not nilpotent and $\varphi$ is injective, neither is $\varphi(h)$. Hence $f^{-1}(D(h)) = D(\varphi(h)) \neq \emptyset$. Thus every open intersects the image of $f$, i.e. $f$ is dominant.

# Exercise 2

(1) $\mathbb{Z}[t]/(p,t) \cong \mathbb{Z}/p\mathbb{Z}$ is a field, so $\mathfrak{m}$ is maximal, i.e. a closed point.

(2) $U = D(p) \cup D(t)$ is clear from $\mathfrak{m}$ being maximal. Hence, $\mathcal{O}_X(U)$ consists of pairs $(s_p, s_t)$ with $s_p \in \mathcal{O}_X(D(p)) = \mathbb{Z}[t, \frac{1}{p}]$ and $s_t \in \mathcal{O}_X(D(t)) = \mathbb{Z}[t, \frac{1}{t}]$ such that their images in $\mathcal{O}_X(D(pt)) = \mathbb{Z}[t, \frac{1}{p}, \frac{1}{t}]$ agree. Clearly, $\mathbb{Z}[t, \frac{1}{p}] \cap \mathbb{Z}[t, \frac{1}{t}] = \mathbb{Z}[t]$, so $\mathcal{O}_X(U) \cong \mathbb{Z}[t]$.

(3) Let $j : U \hookrightarrow X$ be the inclusion. If $U$ were affine, this would be induced by the ring homomorphism $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$, which is an isomorphism by (2). But then $j$ would be an isomorphism of affine schemes, contradicting the fact that $j$ is not surjective.

(4) Since $k[t]$ is a PID, $x$ corresponds to a irreducible polynomial $f$, and $\mathbb{A}^1_k \setminus \{x\} = D(f)$ is affine by the lecture.